

UK Energy Technology Investment Assessment

Project Snapshot

Objective

Assess seven UK energy infrastructure technologies to identify where investment may offer the strongest combination of financial performance, risk, technological attractiveness and decarbonisation potential.

What I analysed

- **7 asset classes**
- Solar PV | Offshore Wind | Battery Storage | Heat Pumps | Hydrogen | Gas Infrastructure | CCUS

4 perspectives

- Technological Attractiveness → Financial → Performance → Risk → Carbon Impact

Key findings

Solar PV - Strong near-term opportunity

9.56% ROI and an Overall Attractiveness Score of 4.

Offshore Wind - Strong investment case

6.77% ROI, low risk score and the highest risk–return score in the analysis.

Battery Storage - Strongest financial performer

Highest average ROI at 9.85%, alongside a strong risk–return position.

Heat Pumps - Attractive, but market constrained

Highest Overall Attractiveness Score, but deployment faces workforce, cost and infrastructure barriers.

Hydrogen - High-potential, longer-term opportunity

Highest estimated carbon savings at 921.80 ktCO₂e, but weaker technology and financial indicators require a more selective approach.

CCUS - Proceed cautiously

Lowest financial performance, with -0.30% ROI, under the current assumptions.

Overall recommendation

Prioritise: Solar PV, Offshore Wind & Battery Storage

Develop selectively: Heat Pumps & Hydrogen

Proceed cautiously: CCUS

Key takeaway: The analysis shows that the most attractive energy investments are not necessarily those with the highest carbon-saving potential. Strong investment decisions require a balanced view of financial return, risk, technology maturity and decarbonisation impact.

Energy Infrastructure Investment Analysis

1. Executive Summary

The Challenge

The transition to net zero is creating a significant need for investment across the energy system. However, capital is constrained, and different energy infrastructure technologies present very different combinations of financial returns, carbon-reduction potential, technological maturity and investment risk.

For investors and decision-makers, the challenge is therefore not simply identifying technologies that can contribute to decarbonisation. **The key question is where investment can deliver the strongest overall value while balancing financial performance and risk against decarbonisation objectives.**

Purpose of the Analysis

This analysis was undertaken to provide a structured comparison of seven energy infrastructure opportunities and assess their relative investment attractiveness.

The analysis considers:

- **Offshore Wind**
- **Solar PV**
- **Gas Infrastructure**
- **Heat Pumps**
- **Battery Storage**
- **CCUS**
- **Hydrogen**

These assets were selected because they represent a broad cross-section of the energy transition, covering renewable generation, electrification, energy storage, existing energy infrastructure and emerging low-carbon technologies. Together, they demonstrate the different investment choices facing organisations as the energy system moves towards net zero.

Rather than assessing technologies purely on their environmental benefits, the analysis combines technology, financial and carbon considerations with investment risk. This allows the assessment to move beyond the question of **"Which technology is greenest?"** towards the more commercially relevant question of:

"Which technologies provide the most attractive investment proposition when financial returns, risk and decarbonisation potential are considered together?"

Overview of Findings

The analysis indicates that **Heat Pumps and Solar PV are the strongest overall performers**, both achieving an Overall Attractiveness Score of 4.

Financially, **Battery Storage records the highest average ROI at 9.85%**, while Offshore Wind combines a 6.77% ROI with a relatively low risk score of 2.

The analysis also highlights an important trade-off between **financial attractiveness and decarbonisation potential**. Hydrogen has exceptionally high estimated carbon savings of approximately **921.80 ktCO₂e**, but its lower technology score means that its significant decarbonisation potential needs to be considered alongside investment and technology risks.

At the other end of the spectrum, CCUS currently presents the weakest investment proposition within the dataset, with an Overall Attractiveness Score of 2, a financial score of 1 and an ROI of -0.30%.

Overall, the findings suggest that **Solar PV, Heat Pumps, Battery Storage and Offshore Wind should receive consideration**, while Hydrogen represents a potentially significant longer-term opportunity and CCUS warrants greater caution under the current assumptions.

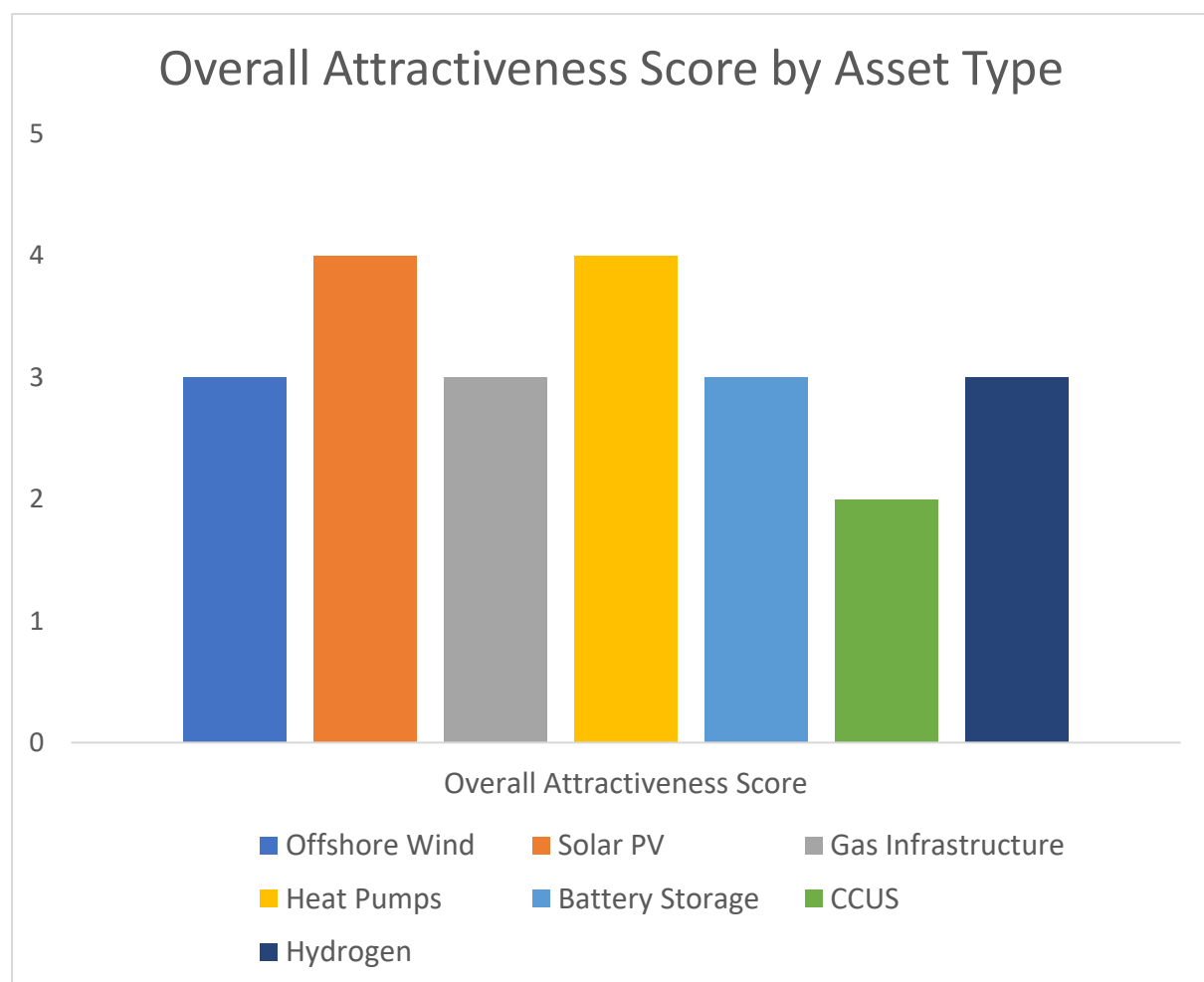
1.1 Methodology & Assumptions

The analysis compares seven energy asset classes using four perspectives: overall attractiveness, financial performance, risk and decarbonisation potential. Financial performance is assessed using ROI and NPV, while carbon savings indicate potential emissions impact. Risk scores reflect planning, grid and supply-chain exposure.

Scores are rated from 1 (least favourable) to 5 (most favourable) based on the underlying criteria and assumptions defined in the analysis. **Please note** that these results should be viewed as a screening assessment, rather than a substitute for detailed investment due diligence.

2. Portfolio Analysis

2.1 Overall Investment Attractiveness



This chart provides the first comparison of the seven energy infrastructure assets based on their overall attractiveness.

The results indicate that **Heat Pumps and Solar PV are the highest-rated assets**, both achieving an Overall Attractiveness Score of 4. This suggests that they provide the strongest balance across the underlying technology, financial and carbon criteria used in the assessment.

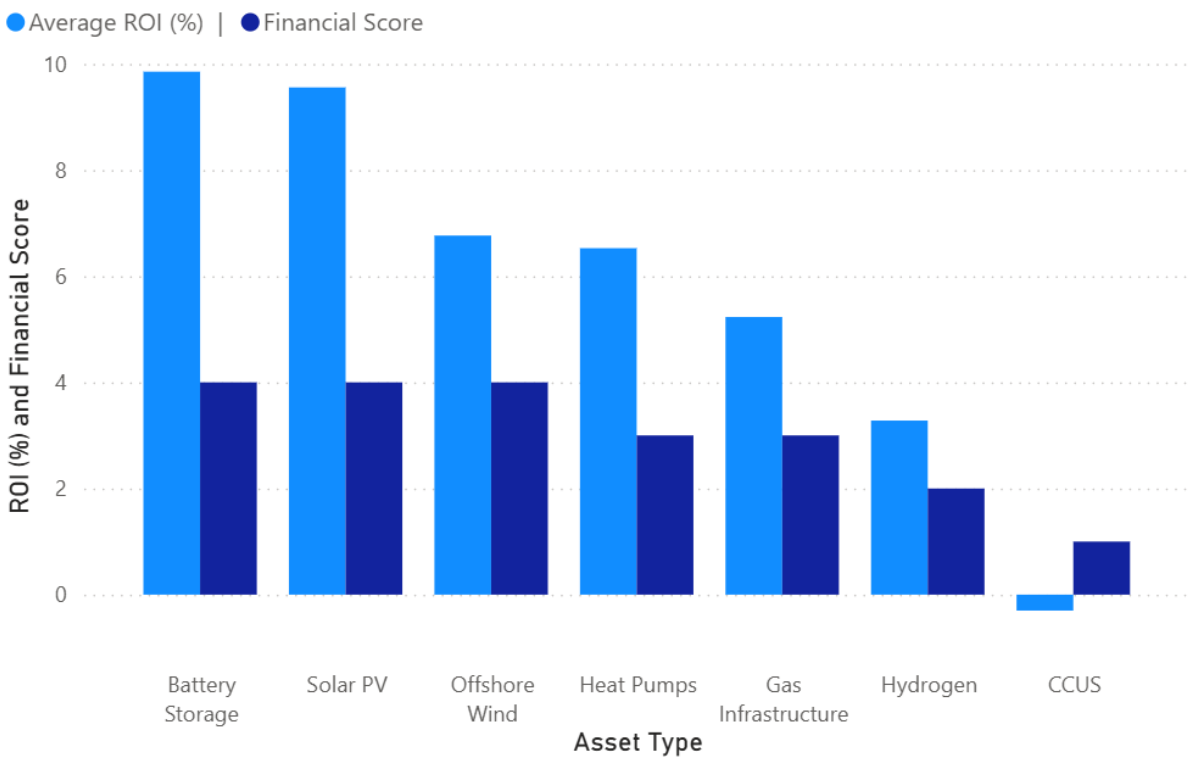
By contrast, **CCUS records the lowest overall score of 2**, indicating a weaker proposition under the assumptions used in this analysis.

The overall score should therefore be viewed as a screening tool rather than a definitive investment decision. It helps identify which assets warrant further investigation before more detailed investment due diligence.

3. Financial Performance

3.1 Financial Attractiveness

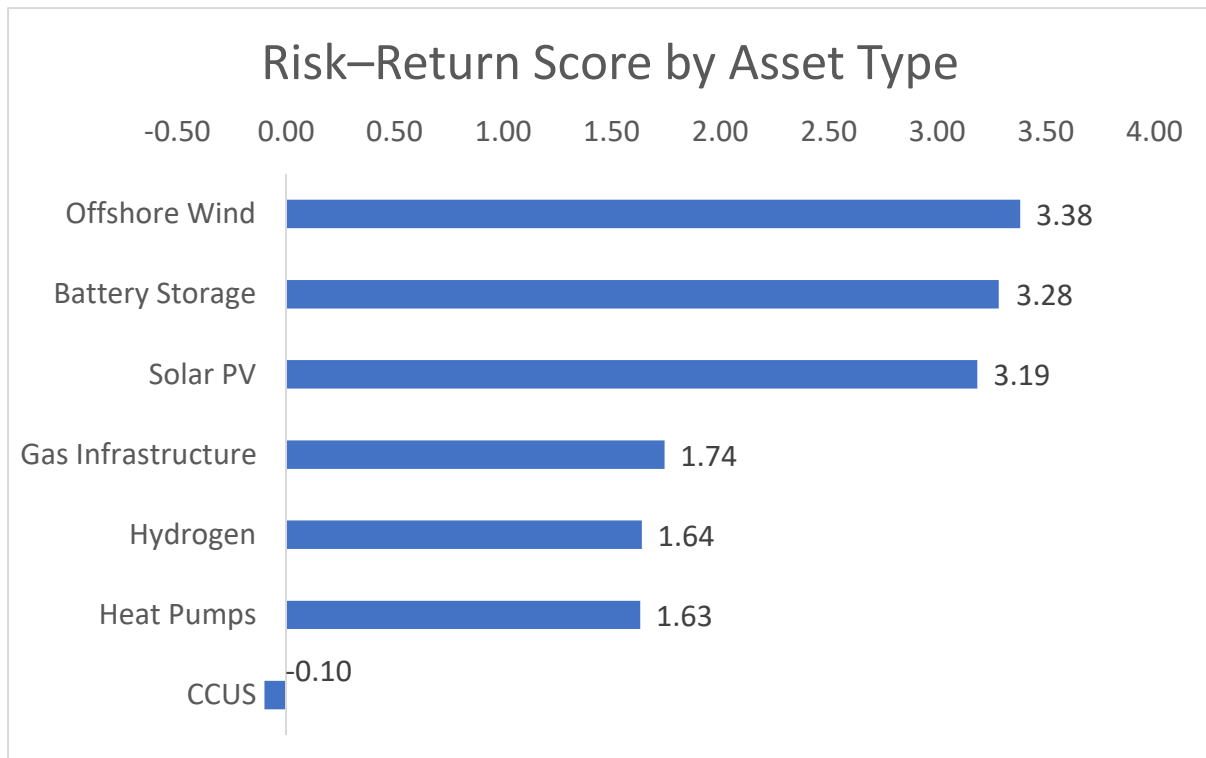
Financial Performance by Asset Type



The financial assessment highlights the differences in commercial potential across the assets. **Solar PV, Offshore Wind and Battery Storage achieve the highest financial scores of 4**, indicating strong financial attractiveness within the scoring framework. Battery Storage also records the highest average ROI at **9.85%**, followed by Solar PV at **9.56%**.

The results show that financial attractiveness varies significantly across the asset classes. While Solar PV, Offshore Wind and Battery Storage achieve the highest financial scores, CCUS records the weakest financial performance, with a negative average ROI. This highlights the importance of considering the financial characteristics of each asset when assessing investment potential.

3.2 Risk–Return Analysis



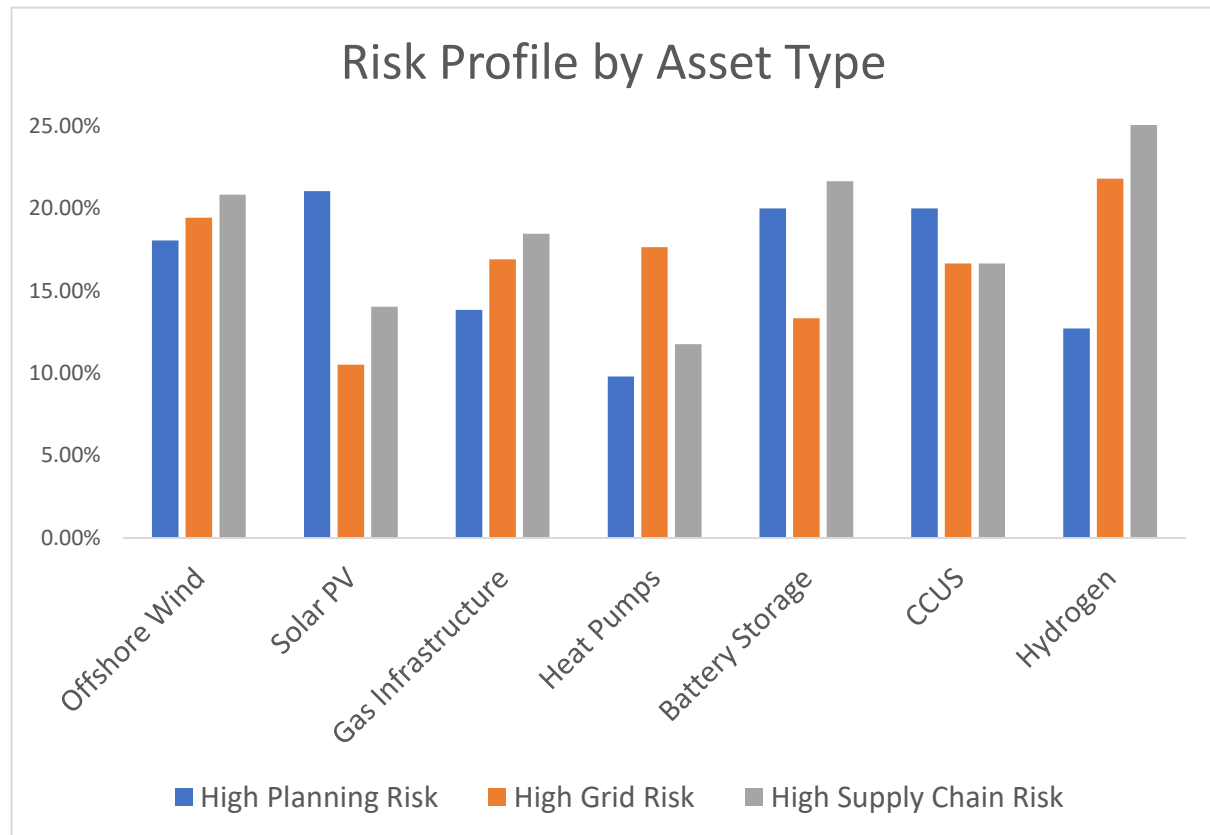
The Risk–Return Score provides an additional perspective by considering the return generated relative to the risk profile assigned to each asset.

The results place Offshore Wind, Battery Storage and Solar PV among the strongest risk–return positions in the analysis. This is particularly useful because an asset with a high absolute ROI may not necessarily be the most attractive once the level of risk is considered.

The analysis therefore highlights the importance of considering **return in the context of risk rather than in isolation**.

4. Risk Analysis

4.1 Risk Profile by Asset

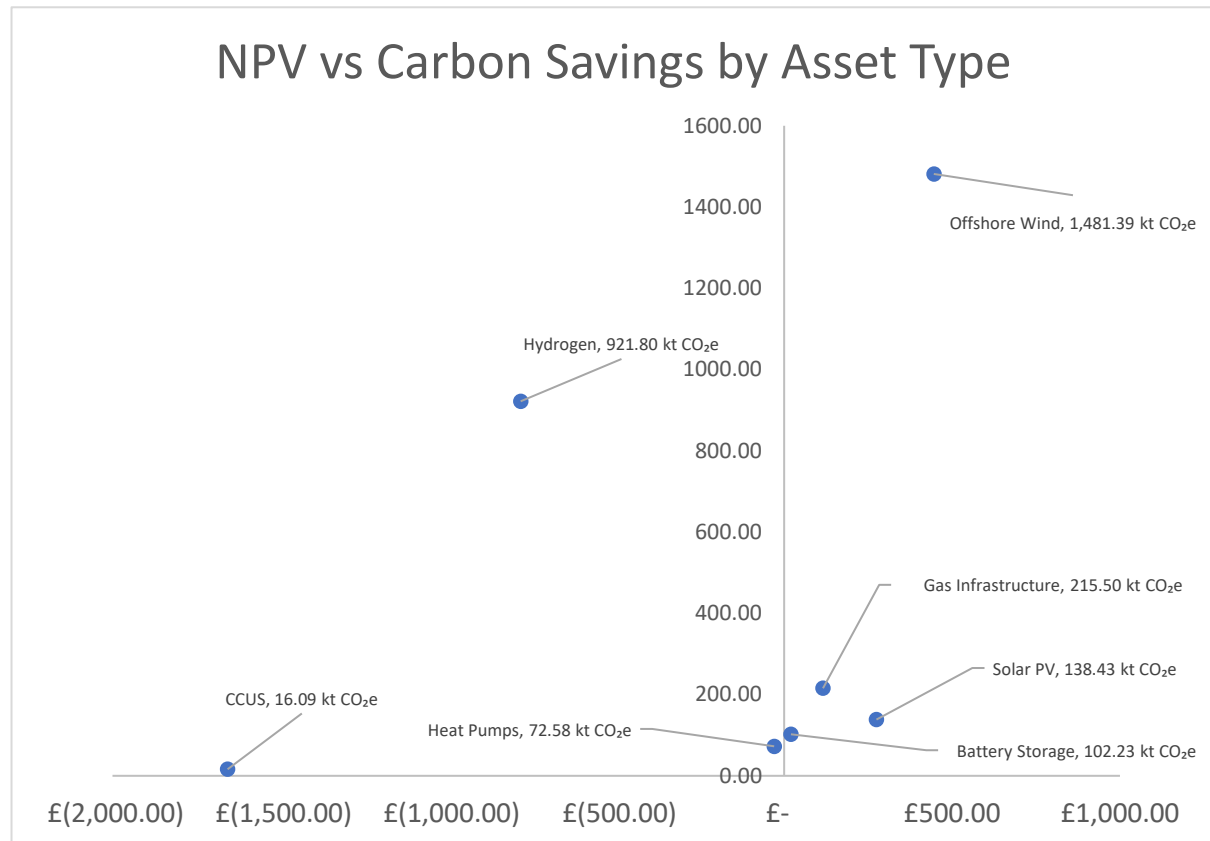


The risk analysis shows that each asset faces a different combination of planning, grid and supply chain risks. These risks are important because they can affect whether a project is developed as planned, how quickly it can become operational, and the potential for delays, cost increases or disruption.

The results show that risk exposure varies significantly across the assets. For example, Hydrogen has relatively high grid and supply chain risk, while Heat Pumps show lower exposure across all three areas. This highlights why risk should be considered alongside financial performance when evaluating investment opportunities. **Higher-risk assets may still be attractive where the potential return is sufficient to compensate investors for taking on additional risk, while lower-risk assets may offer a more stable investment profile.**

5. Decarbonisation Potential

5.1 Carbon Savings



The analysis compares **NPV and estimated carbon savings** to assess both the financial viability and decarbonisation potential of each asset. **NPV was selected because it provides a view of the investment's overall economic value, while carbon savings indicate its potential contribution to emissions reduction.**

The results show significant differences across the assets. Offshore Wind has the highest estimated carbon savings at 1,481.39 ktCO₂e and a positive NPV, indicating strong performance on both financial and decarbonisation measures. Hydrogen also has substantial carbon-saving potential at 921.80 ktCO₂e, but its negative NPV highlights the financial challenges associated with delivering this potential.

This demonstrates that high decarbonisation potential does not necessarily translate into a financially attractive investment. Considering NPV alongside carbon savings therefore helps identify assets that can deliver both economic value and meaningful emissions reductions.

6. Investment Recommendations

Solar PV - Prioritise

The analysis identifies Solar PV as one of the strongest overall opportunities, combining a high attractiveness score with a strong **9.56% ROI and relatively favourable risk profile**. This is also consistent with its established position as a scalable renewable technology, supporting the case for continued investment.

Offshore Wind - Prioritise

Offshore Wind combines strong financial performance with a low-risk score and the highest risk–return score in the analysis. The UK also has an established offshore wind market, with significant investment and a strong project pipeline. Projects such as **Dogger Bank**, including its links to Grimsby and the wider Northeast supply chain, demonstrate the scale of the opportunity. **WindEurope** has previously described the UK as Europe's leading offshore wind market.

Battery Storage - Prioritise

Battery Storage has the highest ROI in the analysis at **9.85% and a strong risk–return score**. Its role in supporting grid flexibility and integrating intermittent renewable generation provides an additional strategic rationale for investment.

Heat Pumps - Attractive, but dependent on market development

Heat Pumps achieve the highest Overall Attractiveness Score but have a lower risk–return position than Solar PV, Offshore Wind and Battery Storage. The technology therefore appears attractive, but deployment remains constrained by factors such as installation and upfront costs, electricity costs, housing suitability and workforce capacity. For example, **Parliament UK** written industry evidence shows that around **1.5 million gas boilers were installed in 2023 compared with only 60–70,000 heat pumps**, while the UK has around 130,000 Gas Safe heating engineers compared with a much smaller heat-pump installation base. **However, schemes such as the Boiler Upgrade Scheme (BUS) and ECO4 are helping to address some of these barriers**, including the upfront cost of heat pumps and energy efficiency improvements in poorly insulated homes.

Hydrogen - Strategic long-term opportunity

Hydrogen has the highest estimated carbon-saving potential in the analysis at **921.80 ktCO₂e**, but its lower technology and risk scores suggest that investment should be approached selectively. This is consistent with the UK market, where significant infrastructure is still being developed. **National Gas' Project Union and FutureGrid programmes**, for example, are being used to develop and test the infrastructure and safety requirements for a future hydrogen transmission network.

CCUS - Selective investment opportunity

CCUS currently ranks weakest financially in the analysis, with a negative ROI and relatively low carbon-saving estimate. However, the technology remains strategically important, particularly for hard-to-abate industries. Major energy companies including **bp, Equinor and TotalEnergies** are already investing in UK carbon capture and storage infrastructure through projects such as the Northern Endurance Partnership and Net Zero Teesside.